

8.1 Charge and Energy

8.1.1 The Continuity Equation

In this chapter we study conservation of energy, momentum, and angular momentum, in electrodynamics. But I want to begin by reviewing the conservation of *charge*, because it is the paradigm for all conservation laws. What precisely does conservation of charge tell us? That the total charge in the Universe is constant? Well, sure – that’s **global** conservation of charge. But **local** conservation of charge is a much stronger statement: if the charge in some region changes, then exactly that amount of charge must have passed in or out through the surface. The tiger can’t simply rematerialize outside the cage; if it got from inside to outside it must have slipped through a hole in the fence.

Formally, the charge in a volume $\mathcal{V}$ is

$$Q(t) = \int_{\mathcal{V}} \rho(\mathbf{r}, t) d\tau, \quad (8.1)$$

and the current flowing out through the boundary $\mathcal{S}$ is $\oint_{\mathcal{S}} \mathbf{J} \cdot d\mathbf{a}$, so local conservation of charge says

$$\frac{dQ}{dt} = - \oint_{\mathcal{S}} \mathbf{J} \cdot d\mathbf{a}. \quad (8.2)$$

Using Eq. 8.1 to rewrite the left-hand side, and invoking the divergence theorem on the right, we have

$$\int_{\mathcal{V}} \frac{\partial \rho}{\partial t} d\tau = - \int_{\mathcal{V}} \nabla \cdot \mathbf{J} d\tau, \quad (8.3)$$

and since this is true for *any* volume, it follows that

$$\boxed{\frac{\partial \rho}{\partial t} = -\nabla \cdot \mathbf{J}.} \quad (8.4)$$

This is the continuity equation – the precise mathematical statement of local conservation of charge. It can be derived from Maxwell’s equation (take a moment to remind yourself how to do that) – conservation of charge is not an *independent* assumption; it is built into the laws of electrodynamics.

It serves as a constraint on the sources (ρ and $\mathbf{J}$). They can't be just *any* old functions – they have to respect conservation of charge.¹

The purpose of this chapter is to develop the corresponding equations for local conservation of energy and momentum. In the process (and perhaps more important) we will learn how to express the energy density and the momentum density (the analogs to ρ), as well as the energy “current” and the momentum “current” (analogous to $\mathbf{J}$).

8.1.2 Poynting's Theorem

In Chapter 2, we found that the work necessary to assemble a static charge distribution (against the Coulomb repulsion of like charges) is (Eq. 2.45)

$$W_e = \frac{\epsilon_0}{2} \int E^2 d\tau,$$

where $\mathbf{E}$ is the resulting electric field. Likewise, the work required to get currents going (against the back emf) is (Eq. 7.44)

$$W_m = \frac{1}{2\mu_0} \int B^2 d\tau,$$

where $\mathbf{B}$ is the resulting magnetic field. This suggests that the total energy stored in electromagnetic fields, per unit volume, is

$$u = \frac{1}{2} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right). \quad (8.5)$$

In this section I will confirm Eq. 8.5, and develop the energy conservation law for electrodynamics.

Suppose we have some charge and current configuration which, at time t , produces fields $\mathbf{E}$ and $\mathbf{B}$. In the next instant, dt , the charges move around a bit. *Question:* How much work, dW , is done by the electromagnetic forces acting on these charges, in the interval dt ? According to the Lorentz force law, the work done on a charge q is

$$dW = \mathbf{F} \cdot d\mathbf{l} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \mathbf{v} dt = q\mathbf{E} \cdot \mathbf{v} dt.$$

In terms of the charge and current *densities*, $q \rightarrow \rho d\tau$ and $\rho\mathbf{v} \rightarrow \mathbf{J}$,² so the rate at which work is done on all the charges in a volume $\mathcal{V}$ is

$$\frac{dW}{dt} = \int_{\mathcal{V}} (\mathbf{E} \cdot \mathbf{J}) d\tau. \quad (8.6)$$

¹ The continuity equation is the *only* such constraint. Any functions $\rho(\mathbf{r}, t)$ and $\mathbf{J}(\mathbf{r}, t)$ consistent with Eq. 8.4 constitute possible charge and current densities, in the sense of admitting solutions to Maxwell's equations.

² Of course, if charges of both signs are present, the *net* charge density can be zero even when the current is *not* – in fact, this is the case for ordinary current-carrying wires. We should really treat the positive and negative charges separately, and combine the two to get Eq. 8.6, with $\mathbf{J} = \rho_+\mathbf{v}_+ + \rho_-\mathbf{v}_-$. See Chapter 5, footnote 8.

Evidently, $\mathbf{E} \cdot \mathbf{J}$ is the work done per unit time, per unit volume – which is to say, the *power* delivered per unit volume. We can express this quantity in terms of the fields alone, using the Ampère–Maxwell law to eliminate $\mathbf{J}$.³

$$\mathbf{E} \cdot \mathbf{J} = \frac{1}{\mu_0} \mathbf{E} \cdot (\nabla \times \mathbf{B}) - \epsilon_0 \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t}.$$

Product rule 6 says

$$\nabla \cdot (\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{B}).$$

Invoking Faraday’s law ($\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$), it follows that

$$\mathbf{E} \cdot (\nabla \times \mathbf{B}) = -\mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t} - \nabla \cdot (\mathbf{E} \times \mathbf{B}).$$

Meanwhile,

$$\mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t} = \frac{1}{2} \frac{\partial}{\partial t} (B^2) \quad \text{and} \quad \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t} = \frac{1}{2} \frac{\partial}{\partial t} (E^2), \quad (8.7)$$

so

$$\mathbf{E} \cdot \mathbf{J} = -\frac{1}{2} \frac{\partial}{\partial t} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) - \frac{1}{\mu_0} \nabla \cdot (\mathbf{E} \times \mathbf{B}). \quad (8.8)$$

Putting this into Eq. 8.6, and applying the divergence theorem to the second term, we have

$$\frac{dW}{dt} = -\frac{d}{dt} \int_{\mathcal{V}} \frac{1}{2} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) d\tau - \frac{1}{\mu_0} \oint_{\mathcal{S}} (\mathbf{E} \times \mathbf{B}) \cdot d\mathbf{a}, \quad (8.9)$$

where $\mathcal{S}$ is the surface bounding $\mathcal{V}$. This is **Poynting’s theorem**; it is the “work-energy theorem” of electrodynamics. The first integral on the right is the total energy stored in the fields, $\int u d\tau$ (Eq. 8.5). The second term evidently represents the rate at which energy is transported out of $\mathcal{V}$, across its boundary surface, by the electromagnetic fields. Poynting’s theorem says, then, that *the work done on the charges by the electromagnetic force is equal to the decrease in energy remaining in the fields, less the energy that flowed out through the surface*.

The *energy per unit time, per unit area*, transported by the fields is called the **Poynting vector**:

$$\mathbf{S} \equiv \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}). \quad (8.10)$$

Specifically, $\mathbf{S} \cdot d\mathbf{a}$ is the energy per unit time crossing the infinitesimal surface $d\mathbf{a}$ – the energy *flux* (so $\mathbf{S}$ is the **energy flux density**).⁴ We will see

³ As you will discover, this is a standard move in field theory: get rid of the *sources* (ρ and $\mathbf{J}$) in favor of the *fields* ($\mathbf{E}$ and $\mathbf{B}$), using the field equations (Gauss and Ampère–Maxwell).

⁴ If you’re very fastidious, you’ll notice a small gap in the logic here: we know from Eq. 8.9 that $\oint \mathbf{S} \cdot d\mathbf{a}$ is the total power passing through a *closed* surface, but this does not prove that $\int \mathbf{S} \cdot d\mathbf{a}$ is the power passing through any *open* surface (there could be an extra term that integrates to zero over all closed surfaces). This is, however, the obvious and natural interpretation; as always, the precise *location* of energy is not really determined in electrodynamics (see Section 2.4.4).

many applications of the Poynting vector in Chapters 9 and 11, but for the moment I am mainly interested in using it to express Poynting's theorem more compactly:

$$\boxed{\frac{dW}{dt} = -\frac{d}{dt} \int_{\mathcal{V}} u d\tau - \oint_S \mathbf{S} \cdot d\mathbf{a}.} \quad (8.11)$$

What if *no* work is done on the charges in $\mathcal{V}$ – what if, for example, $\mathcal{V}$ is in a region of empty space, where there *is* no charge? In that case $dW/dt = 0$, so

$$\int \frac{\partial u}{\partial t} d\tau = -\oint_S \mathbf{S} \cdot d\mathbf{a} = -\int_{\mathcal{V}} (\nabla \cdot \mathbf{S}) d\tau,$$

and hence

$$\frac{\partial u}{\partial t} = -\nabla \cdot \mathbf{S}. \quad (8.12)$$

This is the “continuity equation” for *energy*: u (energy density) plays the role of ρ (charge density), and $\mathbf{S}$ takes the part of $\mathbf{J}$ (current density). It expresses local conservation of electromagnetic energy.

In *general*, though, electromagnetic energy by itself is *not* conserved (nor is the mechanical energy of the charges). Of course not! The fields do work on the charges, and the charges create fields – energy is tossed back and forth between them. In the overall energy economy, you must include the contributions of both the matter and the fields.

Example 8.1. When current flows down a resistive wire, work is done, which shows up as Joule heating of the wire (Eq. 7.7). Though there are certainly *easier* ways to do it, the energy per unit time delivered to the wire can be calculated using the Poynting vector. Assuming it's uniform, the electric field inside the wire is

$$E = \frac{V}{L},$$

where V is the potential difference between the ends and L is the length of the wire (Fig. 8.1). The magnetic field is “circumferential”; at the surface (radius a) it has the value

$$B = \frac{\mu_0 I}{2\pi a}.$$

Accordingly, the magnitude of the Poynting vector is

$$S = \frac{1}{\mu_0} \frac{V}{L} \frac{\mu_0 I}{2\pi a} = \frac{VI}{2\pi aL},$$

and it points radially inward. The energy per unit time passing in through the surface of the wire is therefore

$$\int \mathbf{S} \cdot d\mathbf{a} = S(2\pi aL) = VI,$$

which is exactly what we concluded, on much more direct grounds, in Section 7.1.1.⁵

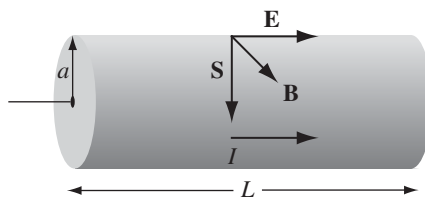


Fig. 8.1

Problem 8.1. Calculate the power (energy per unit time) transported down the cables of Ex. 7.14 and Prob. 7.65, assuming the two conductors are held at potential difference V , and carry current I (down one and back up the other).

Problem 8.2. Imagine two infinite parallel nonconducting planes, a distance d apart, carrying equal and opposite uniform surface charges $\pm\sigma$. A point charge q is at rest in the space between them, a distance z_0 above the $-\sigma$ surface. The electric field is $\mathbf{E} = \mathbf{E}_\sigma + \mathbf{E}_q$. The *total* field energy is infinite, but the **interaction energy** (subtracting off the portions due to σ alone and q alone) is finite.

- Calculate the interaction energy of this configuration.
- Compare the interaction energy to the potential energy of q (take it to be zero at the center between the planes).
- From the gradient of the interaction energy determine the force on q (no surprise).⁶

Problem 8.3. Consider the charging capacitor in Prob. 7.35.

- Find the electric and magnetic fields in the gap, as functions of the distance s from the axis and the time t . (Assume the charge is zero at $t = 0$.)
- Find the energy density u_{em} and the Poynting vector $\mathbf{S}$ in the gap. Note especially the *direction* of $\mathbf{S}$. Check that Eq. 8.12 is satisfied.
- Determine the total energy in the gap, as a function of time. Calculate the total power flowing into the gap, by integrating the Poynting vector over the appropriate surface. Check that the power input is equal to the rate of increase of energy in the gap (Eq. 8.9 – in this case $W = 0$, because there is no charge in the gap). [If you're worried about the fringing fields, do it for a volume of radius $b < a$ well inside the gap.]

⁵ What about energy flow *down* the wire? For a discussion, see M. K. Harbola, *Am. J. Phys.* **78**, 1203 (2010). For a more sophisticated geometry, see B. S. Davis and L. Kaplan, *Am. J. Phys.* **79**, 1155 (2011).

⁶ For commentary, see R. C. Hilborn, *Am. J. Phys.* **84**, 531 (2016).

Problem 8.4. For the charging sphere in Prob. 7.38:

- (a) find the Poynting vector;
- (b) find the energy density in the fields;
- (c) check that the electromagnetic energy is locally conserved (Eq. 8.12), except on the negative z -axis;
- (d) on the negative z -axis, however, external work is being done on the current – pushing the charges in against the electric field. Calculate the external power delivered, and show that it is equal to the rate of increase of the energy in the fields.⁷

8.2 Momentum

8.2.1 Newton's Third Law in Electrodynamics

Imagine a point charge q traveling in along the x -axis at a constant speed v . Because it is moving, its electric field is *not* given by Coulomb's law; nevertheless, $\mathbf{E}$ still points radially outward from the instantaneous position of the charge (Fig. 8.2a), as we'll see in Chapter 10. Since, moreover, a moving point charge does not constitute a steady current, its magnetic field is *not* given by the Biot–Savart law. Nevertheless, it's a fact that $\mathbf{B}$ still circles around the axis in a manner suggested by the right-hand rule (Fig. 8.2b); again, the proof will come in Chapter 10.

Now suppose this charge encounters an identical one, proceeding in at the same speed along the y -axis. Of course, the electromagnetic force between them would tend to drive them off the axes, but let's assume that they're mounted on tracks, or something, so they're obliged to maintain the same direction and the same speed (Fig. 8.3). The electric force between them is repulsive, but how about the magnetic force? Well, the magnetic field of q_1 points into the page (at the position of q_2), so the magnetic force

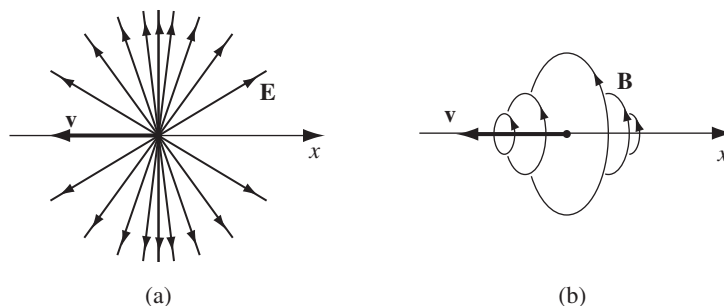


Fig. 8.2

⁷ T. H. Boyer, *Am. J. Phys.* **87**, 729 (2019).

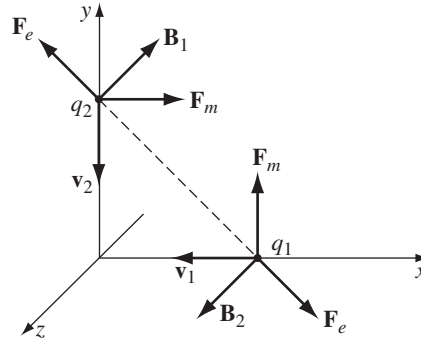


Fig. 8.3

on q_2 is toward the *right*, whereas the magnetic field of q_2 is *out* of the page (at the position of q_1), and the magnetic force on q_1 is *upward*. The net electromagnetic force of q_1 on q_2 is equal but not opposite to the force of q_2 on q_1 , in violation of Newton's third law. In electrostatics and magnetostatics the third law holds, but in electrodynamics it does not.

Well, that's an interesting curiosity, but then, how often does one actually use the third law, in practice? *Answer:* All the time! For the proof of conservation of momentum rests on the cancelation of internal forces, which follows from the third law. When you tamper with the third law, you are placing conservation of momentum in jeopardy, and there is hardly any principle in physics more sacred than *that*.

Momentum conservation is rescued, in electrodynamics, by the realization that the *fields themselves carry momentum*. This is not so surprising when you consider that we have already attributed *energy* to the fields. Whatever momentum is lost to the particles is gained by the fields. Only when the field momentum is added to the mechanical momentum is momentum conservation restored.

8.2.2 Maxwell's Stress Tensor

Let's calculate the total electromagnetic force on the charges in volume $\mathcal{V}$:

$$\mathbf{F} = \int_{\mathcal{V}} (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \rho \, d\tau = \int_{\mathcal{V}} (\rho \mathbf{E} + \mathbf{J} \times \mathbf{B}) \, d\tau. \quad (8.13)$$

The *force per unit volume* is

$$\mathbf{f} = \rho \mathbf{E} + \mathbf{J} \times \mathbf{B}. \quad (8.14)$$

As before, I propose to express this in terms of fields alone, eliminating ρ and $\mathbf{J}$ by using Maxwell's equations (i) and (iv):

$$\mathbf{f} = \epsilon_0 (\nabla \cdot \mathbf{E}) \mathbf{E} + \left(\frac{1}{\mu_0} \nabla \times \mathbf{B} - \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \times \mathbf{B}.$$

Now

$$\frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B}) = \left(\frac{\partial \mathbf{E}}{\partial t} \times \mathbf{B} \right) + \left(\mathbf{E} \times \frac{\partial \mathbf{B}}{\partial t} \right),$$

and Faraday's law says

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E},$$

so

$$\frac{\partial \mathbf{E}}{\partial t} \times \mathbf{B} = \frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B}) + \mathbf{E} \times (\nabla \times \mathbf{E}).$$

Thus

$$\mathbf{f} = \epsilon_0 [(\nabla \cdot \mathbf{E})\mathbf{E} - \mathbf{E} \times (\nabla \times \mathbf{E})] - \frac{1}{\mu_0} [\mathbf{B} \times (\nabla \times \mathbf{B})] - \epsilon_0 \frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B}). \quad (8.15)$$

Just to make things look more symmetrical, let's throw in a term $(\nabla \cdot \mathbf{B})\mathbf{B}$; since $\nabla \cdot \mathbf{B} = 0$, this costs us nothing. Meanwhile, product rule 4 says

$$\nabla(E^2) = 2(\mathbf{E} \cdot \nabla)\mathbf{E} + 2\mathbf{E} \times (\nabla \times \mathbf{E}),$$

so

$$\mathbf{E} \times (\nabla \times \mathbf{E}) = \frac{1}{2} \nabla(E^2) - (\mathbf{E} \cdot \nabla)\mathbf{E},$$

and the same goes for $\mathbf{B}$. Therefore,

$$\begin{aligned} \mathbf{f} = & \epsilon_0 [(\nabla \cdot \mathbf{E})\mathbf{E} + (\mathbf{E} \cdot \nabla)\mathbf{E}] + \frac{1}{\mu_0} [(\nabla \cdot \mathbf{B})\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{B}] \\ & - \frac{1}{2} \nabla(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2) - \epsilon_0 \frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B}). \end{aligned} \quad (8.16)$$

Ugly! But it can be simplified by introducing the **Maxwell stress tensor**,

$$T_{ij} \equiv \epsilon_0 \left(E_i E_j - \frac{1}{2} \delta_{ij} E^2 \right) + \frac{1}{\mu_0} \left(B_i B_j - \frac{1}{2} \delta_{ij} B^2 \right). \quad (8.17)$$

The indices i and j refer to the coordinates x , y , and z , so the stress tensor has a total of nine components (T_{xx} , T_{yy} , T_{zz} , T_{yx} , and so on). The **Kronecker delta**, δ_{ij} , is 1 if the indices are the same ($\delta_{xx} = \delta_{yy} = \delta_{zz} = 1$) and 0 otherwise ($\delta_{xy} = \delta_{xz} = \delta_{yz} = 0$). Thus

$$\begin{aligned} T_{xx} &= \frac{1}{2} \epsilon_0 (E_x^2 - E_y^2 - E_z^2) + \frac{1}{2\mu_0} (B_x^2 - B_y^2 - B_z^2), \\ T_{xy} &= \epsilon_0 (E_x E_y) + \frac{1}{\mu_0} (B_x B_y), \end{aligned}$$

and so on.

Because it carries *two* indices, where a vector has only one, T_{ij} is sometimes written with a double arrow: $\overleftrightarrow{\mathbf{T}}$. One can form the dot product of $\overleftrightarrow{\mathbf{T}}$ with a vector $\mathbf{a}$, in two ways – on the left, and on the right:

$$(\mathbf{a} \cdot \overleftrightarrow{\mathbf{T}})_j = \sum_{i=x,y,z} a_i T_{ij}, \quad (\overleftrightarrow{\mathbf{T}} \cdot \mathbf{a})_j = \sum_{i=x,y,z} T_{ji} a_i. \quad (8.18)$$

The resulting object, which has one remaining index, is itself a vector. In particular, the divergence of $\overleftrightarrow{\mathbf{T}}$ has, as its j th component,

$$(\nabla \cdot \overleftrightarrow{\mathbf{T}})_j = \epsilon_0 \left[(\nabla \cdot \mathbf{E}) E_j + (\mathbf{E} \cdot \nabla) E_j - \frac{1}{2} \nabla_j E^2 \right] \\ + \frac{1}{\mu_0} \left[(\nabla \cdot \mathbf{B}) B_j + (\mathbf{B} \cdot \nabla) B_j - \frac{1}{2} \nabla_j B^2 \right].$$

Thus the force per unit volume (Eq. 8.16) can be written in the much tidier form

$$\mathbf{f} = \nabla \cdot \overleftrightarrow{\mathbf{T}} - \epsilon_0 \mu_0 \frac{\partial \mathbf{S}}{\partial t}, \quad (8.19)$$

where $\mathbf{S}$ is the Poynting vector (Eq. 8.10).

The *total* electromagnetic force on the charges in $\mathcal{V}$ (Eq. 8.13) is

$$\mathbf{F} = \oint_S \overleftrightarrow{\mathbf{T}} \cdot d\mathbf{a} - \epsilon_0 \mu_0 \frac{d}{dt} \int_{\mathcal{V}} \mathbf{S} d\tau. \quad (8.20)$$

(I used the divergence theorem to convert the first term to a surface integral.) In the *static* case the second term drops out, and the electromagnetic force on the charge configuration can be expressed entirely in terms of the stress tensor at the boundary:

$$\mathbf{F} = \oint_S \overleftrightarrow{\mathbf{T}} \cdot d\mathbf{a} \quad (\text{static}). \quad (8.21)$$

Physically, $\overleftrightarrow{\mathbf{T}}$ is the force per unit area (or **stress**) acting on the surface. More precisely, T_{ij} is the force (per unit area) in the i th direction acting on an element of surface oriented in the j th direction – “diagonal” elements (T_{xx} , T_{yy} , T_{zz}) represent **pressure**, and “off-diagonal” elements (T_{xy} , T_{xz} , etc.) are **shears**.

Example 8.2. Determine the net force on the “northern” hemisphere of a (stationary) uniformly charged solid sphere of radius R and charge Q (the same as Prob. 2.48, only this time we’ll use the Maxwell stress tensor and Eq. 8.21).

Solution. The boundary surface consists of two parts – a hemispherical “bowl” at radius R , and a circular disk at $\theta = \pi/2$ (Fig. 8.4). For the bowl,

$$d\mathbf{a} = R^2 \sin \theta d\theta d\phi \hat{\mathbf{r}}$$

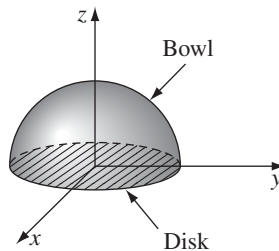


Fig. 8.4

and

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^2} \hat{\mathbf{r}}.$$

In Cartesian components,

$$\hat{\mathbf{r}} = \sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}},$$

so

$$\begin{aligned} T_{zx} &= \epsilon_0 E_z E_x = \epsilon_0 \left(\frac{Q}{4\pi\epsilon_0 R^2} \right)^2 \sin \theta \cos \theta \cos \phi, \\ T_{zy} &= \epsilon_0 E_z E_y = \epsilon_0 \left(\frac{Q}{4\pi\epsilon_0 R^2} \right)^2 \sin \theta \cos \theta \sin \phi, \\ T_{zz} &= \frac{\epsilon_0}{2} (E_z^2 - E_x^2 - E_y^2) = \frac{\epsilon_0}{2} \left(\frac{Q}{4\pi\epsilon_0 R^2} \right)^2 (\cos^2 \theta - \sin^2 \theta). \end{aligned} \quad (8.22)$$

The net force is obviously in the z direction, so it suffices to calculate

$$(\vec{\mathbf{T}} \cdot d\mathbf{a})_z = T_{zx} da_x + T_{zy} da_y + T_{zz} da_z = \frac{\epsilon_0}{2} \left(\frac{Q}{4\pi\epsilon_0 R} \right)^2 \sin \theta \cos \theta d\theta d\phi.$$

The force on the “bowl” is therefore

$$F_{\text{bowl}} = \frac{\epsilon_0}{2} \left(\frac{Q}{4\pi\epsilon_0 R} \right)^2 2\pi \int_0^{\pi/2} \sin \theta \cos \theta d\theta = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{8R^2}. \quad (8.23)$$

Meanwhile, for the equatorial disk,

$$d\mathbf{a} = -r dr d\phi \hat{\mathbf{z}}, \quad (8.24)$$

and (since we are now *inside* the sphere)

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^3} \mathbf{r} = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^3} r (\cos \phi \hat{\mathbf{x}} + \sin \phi \hat{\mathbf{y}}).$$

Thus

$$T_{zz} = \frac{\epsilon_0}{2} (E_z^2 - E_x^2 - E_y^2) = -\frac{\epsilon_0}{2} \left(\frac{Q}{4\pi\epsilon_0 R^3} \right)^2 r^2,$$

and hence

$$(\vec{\mathbf{T}} \cdot d\mathbf{a})_z = \frac{\epsilon_0}{2} \left(\frac{Q}{4\pi\epsilon_0 R^3} \right)^2 r^3 dr d\phi.$$

The force on the disk is therefore

$$F_{\text{disk}} = \frac{\epsilon_0}{2} \left(\frac{Q}{4\pi\epsilon_0 R^3} \right)^2 2\pi \int_0^R r^3 dr = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{16R^2}. \quad (8.25)$$

Combining Eqs. 8.23 and 8.25, I conclude that the net force on the northern hemisphere is

$$F = \frac{1}{4\pi\epsilon_0} \frac{3Q^2}{16R^2}. \quad (8.26)$$

Incidentally, in applying Eq. 8.21, *any* volume that encloses all of the charge in question (and no *other* charge) will do the job. For

example, in the present case we could use the whole region $z > 0$. In that case the boundary surface consists of the entire xy plane (plus a hemisphere at $r = \infty$, but $E = 0$ out there, so it contributes nothing). In place of the “bowl,” we now have the outer portion of the plane ($r > R$). Here,

$$T_{zz} = -\frac{\epsilon_0}{2} \left(\frac{Q}{4\pi\epsilon_0} \right)^2 \frac{1}{r^4}$$

(Eq. 8.22 with $\theta = \pi/2$ and $R \rightarrow r$), and $d\mathbf{a}$ is given by Eq. 8.24, so

$$(\vec{\mathbf{T}} \cdot d\mathbf{a})_z = \frac{\epsilon_0}{2} \left(\frac{Q}{4\pi\epsilon_0} \right)^2 \frac{1}{r^3} dr d\phi,$$

and the contribution from the plane for $r > R$ is

$$\frac{\epsilon_0}{2} \left(\frac{Q}{4\pi\epsilon_0} \right)^2 2\pi \int_R^\infty \frac{1}{r^3} dr = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{8R^2},$$

the same as for the bowl (Eq. 8.23).

I hope you didn’t get too bogged down in the details of Ex. 8.2. If so, take a moment to appreciate what happened. We were calculating the force on a solid object, but instead of doing a *volume* integral, as you might expect, Eq. 8.21 allowed us to set it up as a *surface* integral; somehow the stress tensor sniffs out what is going on inside.

! **Problem 8.5.** Calculate the force of magnetic attraction between the northern and southern hemispheres of a uniformly charged spinning spherical shell, with radius R , angular velocity ω , and surface charge density σ . [This is the same as Prob. 5.47, but this time use the Maxwell stress tensor and Eq. 8.21.]

Problem 8.6.

- (a) Consider two equal point charges q , separated by a distance $2a$. Construct the plane equidistant from the two charges. By integrating Maxwell’s stress tensor over this plane, determine the force of one charge on the other.
- (b) Do the same for charges that are opposite in sign.

8.2.3 Conservation of Momentum

According to Newton’s second law, the force on an object is equal to the rate of change of its momentum:

$$\mathbf{F} = \frac{d\mathbf{p}_{\text{mech}}}{dt}.$$

Equation 8.20 can therefore be written in the form⁸

⁸ Let’s assume the only forces acting are electromagnetic. You can include other forces if you like – both here and in the discussion of energy conservation – but they are just a distraction from the essential story.

$$\frac{d\mathbf{p}_{\text{mech}}}{dt} = -\epsilon_0\mu_0 \frac{d}{dt} \int_{\mathcal{V}} \mathbf{S} d\tau + \oint_S \overleftrightarrow{\mathbf{T}} \cdot d\mathbf{a}, \quad (8.27)$$

where $\mathbf{p}_{\text{mech}}$ is the (mechanical) momentum ($\sum m_i \mathbf{v}_i$) of the particles in volume $\mathcal{V}$.⁹ This expression is similar in structure to Poynting's theorem (Eq. 8.11), and it invites an analogous interpretation: the first integral represents *momentum stored in the fields*,

$$\mathbf{p} = \mu_0\epsilon_0 \int_{\mathcal{V}} \mathbf{S} d\tau, \quad (8.28)$$

while the second integral is the *momentum per unit time flowing in through the surface*.

Equation 8.27 is the statement of *conservation of momentum* in electrodynamics: if the mechanical momentum increases, either the field momentum decreases, or else the fields are carrying momentum into the volume through the surface. The momentum *density* in the fields is evidently

$$\mathbf{g} = \mu_0\epsilon_0 \mathbf{S} = \epsilon_0(\mathbf{E} \times \mathbf{B}), \quad (8.29)$$

and the momentum flux transported by the fields is $-\overleftrightarrow{\mathbf{T}}$ (more precisely, $-\overleftrightarrow{\mathbf{T}} \cdot d\mathbf{a}$ is the electromagnetic momentum per unit time passing through the area $d\mathbf{a}$).

If the mechanical momentum in $\mathcal{V}$ is not changing (for example, if we are talking about a region of empty space), then

$$\int \frac{\partial \mathbf{g}}{\partial t} d\tau = \oint \overleftrightarrow{\mathbf{T}} \cdot d\mathbf{a} = \int \nabla \cdot \overleftrightarrow{\mathbf{T}} d\tau,$$

and hence

$$\frac{\partial \mathbf{g}}{\partial t} = \nabla \cdot \overleftrightarrow{\mathbf{T}}. \quad (8.30)$$

This is the “continuity equation” for electromagnetic momentum, with $\mathbf{g}$ (momentum density) in the role of ρ (charge density) and $-\overleftrightarrow{\mathbf{T}}$ playing the part of $\mathbf{J}$. Equation 8.30 expresses the local conservation of field momentum. But in general (when there *are* charges around) the field momentum by itself, and the mechanical momentum by itself, are *not* conserved – charges and fields exchange momentum, and only the *total* is conserved.

Notice that the Poynting vector has appeared in two quite different roles: $\mathbf{S}$ itself is the energy per unit area, per unit time, *transported* by the electromagnetic fields, while $\mu_0\epsilon_0 \mathbf{S}$ is the momentum per unit volume *stored* in those fields.¹⁰ Similarly, $\overleftrightarrow{\mathbf{T}}$ plays a dual role: $\overleftrightarrow{\mathbf{T}}$ itself is the electromagnetic stress (force per unit area) acting on a surface, and $-\overleftrightarrow{\mathbf{T}}$ describes the flow of momentum (it is the momentum current density) carried by the fields.

⁹ Again, to avoid distracting complications I assume no particles enter or leave $\mathcal{V}$.

¹⁰ This is no coincidence – see R. P. Feynman, R. B. Leighton, and M. Sands, *The Feynman Lectures on Physics*, Vol. 2, Addison-Wesley, Reading, MA (1964), Sect. 27-6.

Example 8.3. A long resistanceless coaxial cable, of length l , consists of an inner conductor (radius a) and an outer conductor (radius b). It is connected to a battery at one end and a resistor at the other (Fig. 8.5). The inner conductor carries a uniform charge per unit length λ , and a steady current I to the right; the outer conductor has the opposite charge and current. What is the electromagnetic momentum stored in the fields?

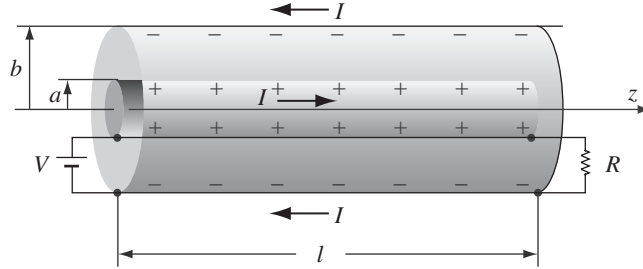


Fig. 8.5

Solution. The fields are

$$\mathbf{E} = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{s} \hat{\mathbf{s}}, \quad \mathbf{B} = \frac{\mu_0}{2\pi} \frac{I}{s} \hat{\boldsymbol{\phi}}.$$

The Poynting vector is therefore

$$\mathbf{S} = \frac{\lambda I}{4\pi^2 \epsilon_0 s^2} \hat{\mathbf{z}}.$$

So energy is flowing down the line, from the battery to the resistor. In fact, the power transported is

$$P = \int \mathbf{S} \cdot d\mathbf{a} = \frac{\lambda I}{4\pi^2 \epsilon_0} \int_a^b \frac{1}{s^2} 2\pi s ds = \frac{\lambda I}{2\pi \epsilon_0} \ln(b/a) = IV,$$

as it should be.

The *momentum* in the fields is

$$\mathbf{p} = \mu_0 \epsilon_0 \int \mathbf{S} d\tau = \frac{\mu_0 \lambda I}{4\pi^2} \hat{\mathbf{z}} \int_a^b \frac{1}{s^2} l 2\pi s ds = \frac{\mu_0 \lambda I l}{2\pi} \ln(b/a) \hat{\mathbf{z}} = \epsilon_0 \mu_0 IV l \hat{\mathbf{z}}.$$

This is an astonishing result. The cable is not moving, $\mathbf{E}$ and $\mathbf{B}$ are static, and yet we are asked to believe that there is momentum in the fields. If something tells you this cannot be the whole story, you have sound intuitions. But the resolution of this paradox will have to await Chapter 12 (Ex. 12.12).

Suppose now that we turn up the resistance, so the current decreases. The changing magnetic field will induce an electric field (Eq. 7.29):

$$\mathbf{E} = \left[\frac{\mu_0}{2\pi} \frac{dI}{dt} \ln s + K \right] \hat{\mathbf{z}}.$$

This field exerts a force on $\pm\lambda$:

$$\mathbf{F} = \lambda l \left[\frac{\mu_0}{2\pi} \frac{dI}{dt} \ln a + K \right] \hat{\mathbf{z}} - \lambda l \left[\frac{\mu_0}{2\pi} \frac{dI}{dt} \ln b + K \right] \hat{\mathbf{z}} = -\frac{\mu_0 \lambda l}{2\pi} \frac{dI}{dt} \ln(b/a) \hat{\mathbf{z}}.$$

The total momentum imparted to the cable, as the current drops from I to 0, is therefore

$$\mathbf{p}_{\text{mech}} = \int \mathbf{F} dt = \frac{\mu_0 \lambda I l}{2\pi} \ln(b/a) \hat{\mathbf{z}},$$

which is precisely the momentum originally stored in the fields.

Problem 8.7. Imagine two parallel infinite sheets, carrying uniform surface charge $+\sigma$ (on the sheet at $z = d$) and $-\sigma$ (at $z = 0$). They are moving in the y direction at constant speed v (as in Prob. 5.18).

- What is the electromagnetic momentum in a region of area A ?
- Now suppose the top sheet moves slowly down (speed u) until it reaches the bottom sheet, so the fields disappear. By calculating the total force on the charge ($q = \sigma A$), show that the impulse delivered to the sheet is equal to the momentum originally stored in the fields. [Hint: As the upper plate passes by, the magnetic field drops to zero, inducing an electric field that delivers an impulse to the lower plate.]

Problem 8.8. A charged parallel-plate capacitor (with uniform electric field $\mathbf{E} = E\hat{\mathbf{z}}$) is placed in a uniform magnetic field $\mathbf{B} = B\hat{\mathbf{x}}$, as shown in Fig. 8.6.

- Find the electromagnetic momentum in the space between the plates.
- Now a resistive wire is connected between the plates, along the z -axis, so that the capacitor slowly discharges. The current through the wire will experience a magnetic force; what is the total impulse delivered to the system, during the discharge?¹¹

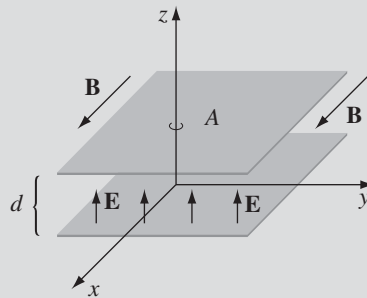


Fig. 8.6

Problem 8.9. Consider an infinite parallel-plate capacitor, with the lower plate (at $z = -d/2$) carrying surface charge density $-\sigma$, and the upper plate (at $z = +d/2$) carrying charge density $+\sigma$.

¹¹ There is *much* more to be said about this problem, so don't get too excited if your answers to (a) and (b) appear to be consistent. See D. Babson et al., *Am. J. Phys.* **77**, 826 (2009).

- (a) Determine all nine elements of the stress tensor, in the region between the plates. Display your answer as a 3×3 matrix:

$$\begin{pmatrix} T_{xx} & T_{xy} & T_{xz} \\ T_{yx} & T_{yy} & T_{yz} \\ T_{zx} & T_{zy} & T_{zz} \end{pmatrix}.$$

- (b) Use Eq. 8.21 to determine the electromagnetic force per unit area on the top plate. Compare Eq. 2.51.
- (c) What is the electromagnetic momentum per unit area, per unit time, crossing the xy plane (or any other plane parallel to that one, between the plates)?
- (d) Of course, there must be *mechanical* forces holding the plates apart – perhaps the capacitor is filled with insulating material under pressure. Suppose we suddenly *remove* the insulator; the momentum flux (c) is now absorbed by the plates, and they begin to move. Find the momentum per unit time delivered to the top plate (which is to say, the force acting on it) and compare your answer to (b). [Note: This is not an *additional* force, but rather an alternative way of calculating the *same* force – in (b) we got it from the force law, and in (d) we do it by conservation of momentum.]

8.2.4 Angular Momentum

By now, the electromagnetic fields (which started out as mediators of forces between charges) have taken on a life of their own. They carry *energy* (Eq. 8.5)

$$u = \frac{1}{2} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right), \quad (8.31)$$

and *momentum* (Eq. 8.29)

$$\mathbf{g} = \epsilon_0 (\mathbf{E} \times \mathbf{B}), \quad (8.32)$$

and, for that matter, *angular* momentum

$$\boldsymbol{\ell} = \mathbf{r} \times \mathbf{g} = \epsilon_0 [\mathbf{r} \times (\mathbf{E} \times \mathbf{B})]. \quad (8.33)$$

Even perfectly *static* fields can harbor momentum and angular momentum, as long as $\mathbf{E} \times \mathbf{B}$ is nonzero, and it is only when these field contributions are included that the conservation laws are sustained.

Example 8.4. Imagine a very long solenoid with radius R , n turns per unit length, and current I . Coaxial with the solenoid are two long cylindrical (nonconducting) shells of length l – one, *inside* the solenoid at radius a , carries a charge $+Q$, uniformly distributed over its surface; the other, *outside* the solenoid at radius b , carries charge $-Q$ (see Fig. 8.7; l is supposed to be much greater than b). When the current in the solenoid is gradually reduced, the cylinders begin to rotate, as

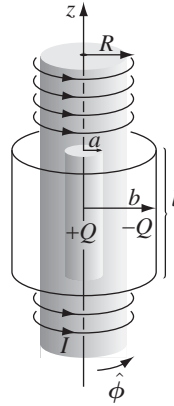


Fig. 8.7

we found in Ex. 7.9. *Question:* Where does the angular momentum come from?¹²

Solution. It was initially stored in the fields. Before the current was switched off, there was an electric field,

$$\mathbf{E} = \frac{Q}{2\pi\epsilon_0 l} \frac{1}{s} \hat{\mathbf{s}} \quad (a < s < b),$$

in the region between the cylinders, and a magnetic field,

$$\mathbf{B} = \mu_0 n I \hat{\mathbf{z}} \quad (s < R),$$

inside the solenoid. The momentum density (Eq. 8.29) was therefore

$$\mathbf{g} = -\frac{\mu_0 n I Q}{2\pi l s} \hat{\boldsymbol{\phi}},$$

in the region $a < s < R$. The z component of the *angular* momentum density was

$$(\mathbf{r} \times \mathbf{g})_z = -\frac{\mu_0 n I Q}{2\pi l},$$

which is *constant* (independent of s). To get the *total* angular momentum in the fields, we simply multiply by the volume, $\pi(R^2 - a^2)l$:¹³

$$\mathbf{L} = -\frac{1}{2} \mu_0 n I Q (R^2 - a^2) \hat{\mathbf{z}}. \quad (8.34)$$

When the current is turned off, the changing magnetic field induces a circumferential electric field, given by Faraday's law:

¹² This is a variation on the “Feynman disk paradox” (R. P. Feynman, R. B. Leighton, and M. Sands, *The Feynman Lectures on Physics*, Vol. 2, Addison-Wesley, Reading, MA (1964), pp. 17-5) suggested by F. L. Boos, Jr., *Am. J. Phys.* **52**, 756 (1984). A similar model was proposed earlier by R. H. Romer, *Am. J. Phys.* **34**, 772 (1966). For further references, see T.-C. E. Ma, *Am. J. Phys.* **54**, 949 (1986).

¹³ The radial component integrates to zero, by symmetry.

$$\mathbf{E} = \begin{cases} -\frac{1}{2}\mu_0 n \frac{dI}{dt} \frac{R^2}{s} \hat{\phi} & (s > R), \\ -\frac{1}{2}\mu_0 n \frac{dI}{dt} s \hat{\phi} & (s < R). \end{cases}$$

Thus the torque on the outer cylinder is

$$\mathbf{N}_b = \mathbf{r} \times (-Q\mathbf{E}) = \frac{1}{2}\mu_0 n Q R^2 \frac{dI}{dt} \hat{\mathbf{z}},$$

and it picks up an angular momentum

$$\mathbf{L}_b = \frac{1}{2}\mu_0 n Q R^2 \hat{\mathbf{z}} \int_I^0 \frac{dI}{dt} dt = -\frac{1}{2}\mu_0 n I Q R^2 \hat{\mathbf{z}}.$$

Similarly, the torque on the inner cylinder is

$$\mathbf{N}_a = -\frac{1}{2}\mu_0 n Q a^2 \frac{dI}{dt} \hat{\mathbf{z}},$$

and its angular momentum increase is

$$\mathbf{L}_a = \frac{1}{2}\mu_0 n I Q a^2 \hat{\mathbf{z}}.$$

So it all works out: $\mathbf{L}_{\text{em}} = \mathbf{L}_a + \mathbf{L}_b$. The angular momentum *lost* by the fields is precisely equal to the angular momentum *gained* by the cylinders, and the *total* angular momentum (fields plus matter) is conserved.

Caution: Field energy does *not* obey the superposition principle – the energy of a composite system is *not* the sum of the energies of its constituent parts. That's because energy (Eq. 8.5) is *quadratic* in $\mathbf{E}$ and $\mathbf{B}$. The fields themselves *do* satisfy the superposition principle, so, for instance,

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2,$$

and hence the electric part of the energy density is

$$u = \frac{1}{2}\epsilon_0(\mathbf{E} \cdot \mathbf{E}) = \frac{1}{2}\epsilon_0(E_1^2 + E_2^2 + 2\mathbf{E}_1 \cdot \mathbf{E}_2) = u_1 + u_2 + \epsilon_0\mathbf{E}_1 \cdot \mathbf{E}_2.$$

The cross term spoils it. To take a trivial example, imagine a sphere of charge Q ; the same sphere with double the charge would have twice the *field*, but *four* times the energy. And the same goes for the Poynting vector, momentum, angular momentum, and the stress tensor, since these are all quadratic in the fields.

Problem 8.10. In Ex. 8.4, suppose that instead of turning off the *magnetic* field (by reducing I) we turn off the *electric* field, by connecting a weakly¹⁴ conducting radial spoke between the cylinders. (We'll have to cut a slot in the solenoid, so the cylinders can still rotate freely.) From the magnetic force on the current in the spoke, determine the total angular momentum

¹⁴ In Ex. 8.4 we turned the current off slowly, to keep things quasistatic; here we reduce the electric field slowly to keep the displacement current negligible.

delivered to the cylinders, as they discharge (they are now rigidly connected, so they rotate together). Compare the initial angular momentum stored in the fields (Eq. 8.34). (Notice that the *mechanism* by which angular momentum is transferred from the fields to the cylinders is entirely different in the two cases: in Ex. 8.4 it was Faraday's law, but here it is the Lorentz force law.)

Problem 8.11. Two concentric spherical shells carry uniformly distributed charges $+Q$ (at radius a) and $-Q$ (at radius $b > a$). They are immersed in a uniform magnetic field $\mathbf{B} = B_0 \hat{\mathbf{z}}$.

- (a) Find the angular momentum of the fields (with respect to the center).
- (b) Now the magnetic field is gradually turned off. Find the torque on each sphere, and the resulting angular momentum of the system.

! **Problem 8.12.**¹⁵ Imagine an iron sphere of radius R that carries a charge Q and a uniform magnetization $\mathbf{M} = M \hat{\mathbf{z}}$. The sphere is initially at rest.

- (a) Compute the angular momentum stored in the electromagnetic fields.
- (b) Suppose the sphere is gradually (and uniformly) demagnetized (perhaps by heating it up past the Curie point). Use Faraday's law to determine the induced electric field, find the torque this field exerts on the sphere, and calculate the total angular momentum imparted to the sphere in the course of the demagnetization.
- (c) Suppose instead of *demagnetizing* the sphere we *discharge* it, by connecting a grounding wire to the north pole. Assume the current flows over the surface in such a way that the (surface) charge density remains uniform. Use the Lorentz force law to determine the torque on the sphere, and calculate the total angular momentum imparted to the sphere in the course of the discharge. (The magnetic field is discontinuous at the surface ... does this matter?) [Answer: $\frac{2}{9} \mu_0 M Q R^2$.]

8.3 Magnetic Forces Do No Work

This¹⁶ is perhaps a good place to revisit the old paradox that magnetic forces do no work (Eq. 5.11). What about that magnetic crane lifting the carcass of a junked car? *Somebody* is doing work on the car, and if it's not the magnetic field, who *is* it? The car is ferromagnetic; in the presence of the magnetic field, it contains a lot of microscopic magnetic dipoles (spinning electrons, actually), all lined up. The resulting magnetization is equivalent to a bound current running around the surface, so let's model the car as a circular current loop.

¹⁵ This version of the Feynman disk paradox was proposed by N. L. Sharma, *Am. J. Phys.* **56**, 420 (1988); similar models were analyzed by E. M. Pugh and G. E. Pugh, *Am. J. Phys.* **35**, 153 (1967) and by R. H. Romer, *Am. J. Phys.* **35**, 445 (1967).

¹⁶ This section can be skipped without loss of continuity. I include it for those readers who are disturbed by the notion that magnetic forces do no work.

Example 8.5. Picture the current loop as a nonconducting ring, of line charge λ , rotating at angular velocity ω (Fig. 8.8). The upward magnetic force on the loop is (Eq. 6.2)

$$F = 2\pi I a B_s, \quad (8.35)$$

where B_s is the radial component of the magnet's field,¹⁷ and $I = \lambda \omega a$. If the ring rises a distance dz (while the magnet itself stays put), the work done on it is

$$dW = 2\pi a^2 \lambda \omega B_s dz. \quad (8.36)$$

This increases the potential energy of the ring. Who did the work? Naïvely, it appears that the magnetic field is responsible, but we have already learned (Ex. 5.3) that this is not the case – as the ring rises, the magnetic force is perpendicular to the *net* velocity of the charges in the ring, so it does *no* work on them.

At the same time, however, a motional emf is induced in the ring, which opposes the flow of charge, and hence reduces its angular velocity:

$$\mathcal{E} = -\frac{d\Phi}{dt}.$$

Here $d\Phi$ is the flux through the “ribbon” joining the ring at time t to the ring at time $t + dt$ (Fig. 8.9):

$$d\Phi = B_s 2\pi a dz.$$

Now

$$\mathcal{E} = \oint \mathbf{f} \cdot d\mathbf{l} = f(2\pi a),$$

where $\mathbf{f}$ is the force per unit charge. So

$$f = -B_s \frac{dz}{dt}, \quad (8.37)$$

the force on a segment of length dl is $f \lambda dl$, the torque on the ring is

$$N = a \left(-B_s \frac{dz}{dt} \right) \lambda (2\pi a),$$

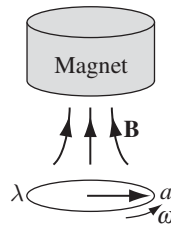


Fig. 8.8

¹⁷ Note that the field has to be *nonuniform*, or it won't lift the car at all. Also, stipulate for now that the magnetic field remains *constant* as the loop rises.



Fig. 8.9

and the work done (slowing the rotation) is $N d\phi = N \omega dt$, or

$$dW = -2\pi a^2 \lambda \omega B_s dz. \quad (8.38)$$

The ring slows down, and the rotational energy it loses (Eq. 8.38) is precisely equal to the potential energy it gains (Eq. 8.36). All the magnetic field did was convert energy from one form to another. If you'll permit some sloppy language, the work done by the vertical *component* of the magnetic force (Eq. 8.35) is equal and opposite to the work done by its horizontal *component* (Eq. 8.37).¹⁸

Example 8.6. What about the magnet? Is it completely passive in this process? Suppose we model it as a big circular loop (radius b), resting on a table and carrying a current I_b ; the “junk car” is a relatively small current loop (radius a), on the floor directly below, carrying a current I_a (Fig. 8.10). This time, just for a change, let's assume both currents are constant (we'll include a regulated power supply in each loop¹⁹). Parallel currents attract, and we propose to lift the small loop off the floor, keeping careful track of the work done and the agency responsible.

Let's start by adjusting the currents so the small ring just “floats,” a distance h below the table, with the magnetic force exactly balancing the weight ($m_a g$) of the little ring. I'll let you calculate the magnetic force (Prob. 8.13):

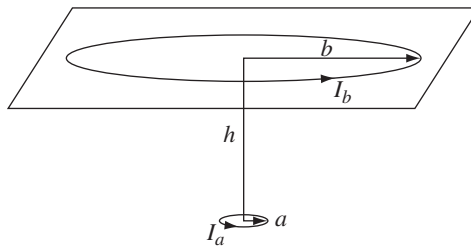


Fig. 8.10

¹⁸ This argument is essentially the same as the one in Ex. 5.3, except that in this case I told the story in terms of motional emf, instead of the Lorentz force law. But after all, the flux rule is a *consequence* of the Lorentz force law.

¹⁹ The lower loop could be a single spinning electron, in which case quantum mechanics fixes its angular momentum at $\hbar/2$. It might appear that this sustains the current, with no need for a power supply. I will return to this point, but for now let's just keep quantum mechanics out of it.

$$F_{\text{mag}} = \frac{3\pi}{2} \mu_0 I_a I_b \frac{a^2 b^2 h}{(b^2 + h^2)^{5/2}} = m_a g. \quad (8.39)$$

Now the loop rises an infinitesimal distance dz ; the work done is equal to the gain in its potential energy

$$dW_g = m_a g dz = \frac{3\pi}{2} \mu_0 I_a I_b \frac{a^2 b^2 h}{(b^2 + h^2)^{5/2}} dz. \quad (8.40)$$

Who did it? The magnetic field? *No!* The work was done by the power supply that sustains the current in loop a (Ex. 5.3). As the loop rises, a motional emf is induced in it. The flux through the loop is

$$\Phi_a = M I_b,$$

where M is the mutual inductance of the two loops:

$$M = \frac{\pi \mu_0}{2} \frac{a^2 b^2}{(b^2 + h^2)^{3/2}}$$

(Prob. 7.23). The emf is

$$\begin{aligned} \mathcal{E}_a &= -\frac{d\Phi_a}{dt} = -I_b \frac{dM}{dt} = -I_b \frac{dM}{dh} \frac{dh}{dt} \\ &= -I_b \left(-\frac{3}{2}\right) \frac{\pi \mu_0}{2} \frac{a^2 b^2}{(b^2 + h^2)^{5/2}} 2h \frac{(-dz)}{dt}. \end{aligned}$$

The work done by the power supply (fighting against this motional emf) is

$$dW_a = -\mathcal{E}_a I_a dt = \frac{3\pi}{2} \mu_0 I_a I_b \frac{a^2 b^2 h}{(b^2 + h^2)^{5/2}} dz \quad (8.41)$$

– the same as the work done in lifting the loop (Eq. 8.40).

Meanwhile, however, a *Faraday* emf is induced in the *upper* loop, due to the changing flux from the lower loop:

$$\Phi_b = M I_a \Rightarrow \mathcal{E}_b = -I_a \frac{dM}{dt},$$

and the work done by the power supply in ring b (to sustain the current I_b) is

$$dW_b = -\mathcal{E}_b I_b dt = \frac{3\pi}{2} \mu_0 I_a I_b \frac{a^2 b^2 h}{(b^2 + h^2)^{5/2}} dz, \quad (8.42)$$

exactly the same as dW_a . That's embarrassing – the power supplies have done *twice* as much work as was necessary to lift the junk car! Where did the “wasted” energy go? *Answer:* It increased the energy stored in the fields. The energy in a system of two current-carrying loops is (see Prob. 8.14)

$$U = \frac{1}{2} L_a I_a^2 + \frac{1}{2} L_b I_b^2 + M I_a I_b, \quad (8.43)$$

so

$$dU = I_a I_b \frac{dM}{dt} dt = dW_b.$$

Remarkably, all four energy increments are the same. If we care to apportion things this way, the power supply in loop *a* contributes the energy necessary to lift the lower ring, while the power supply in loop *b* provides the extra energy for the fields. If all we're interested in is the work done to raise the ring, we can ignore the upper loop (and the energy in the fields) altogether.

Example 8.7. In both these models, the magnet itself was stationary. That's like lifting a paper clip by holding a magnet over it. But in the case of the magnetic crane, the car stays in contact with the magnet, which is attached to a cable that lifts the whole works. As a model, we might stick the upper loop in a big box, the lower loop in a little box, and crank up the currents so the force of attraction is much greater than m_ag ; the two boxes snap together, and we attach a string to the upper box and pull up on it (Fig. 8.11).

The same old mechanism (Ex. 5.3) prevails: as the lower loop rises, the magnetic force tilts backward; its vertical component lifts the loop, but its horizontal component opposes the current, and no net work is done. This time, however, the motional emf is perfectly balanced by the Faraday emf fighting to keep the current going – the flux through the lower loop is not changing. (If you like, the flux is *increasing* because the loop is moving upward, into a region of higher magnetic field, but it is *decreasing* because the magnetic field of the upper loop – at any given point in space – is decreasing as that loop moves up. See Prob. 7.69.) No power supply is needed to sustain the current (and for that matter, no power supply is required in the upper loop either, since the energy in the fields is not changing). Who did the work to lift the car? The person pulling up on the rope, obviously. The role of the magnetic field was merely to transmit this energy to the car, via the vertical component of the magnetic force. But the magnetic field itself (as always) did no work.

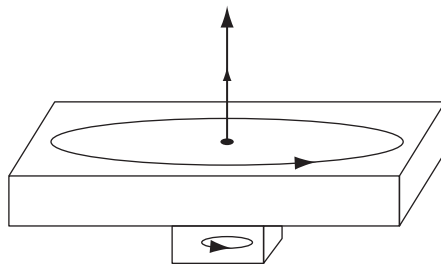


Fig. 8.11

The fact that magnetic forces do no work follows directly from the Lorentz force law, so if you think you have discovered an exception, you're going to have to explain why that law is incorrect. For example, if magnetic

monopoles exist, the force on a particle with electric charge q_e and magnetic charge q_m becomes (Prob. 7.40)

$$\mathbf{F} = q_e(\mathbf{E} + \mathbf{v} \times \mathbf{B}) + q_m(\mathbf{B} - \epsilon_0\mu_0\mathbf{v} \times \mathbf{E}). \quad (8.44)$$

In that case, magnetic fields *can* do work ... but *only on magnetic charges*. So unless your car is made of monopoles (I don't think so), this doesn't solve the problem.

A somewhat less radical possibility is that in addition to electric charges there exist permanent point magnetic dipoles (electrons?), whose dipole moment $\mathbf{m}_i$ is not associated with any electric current, but simply *is*. The Lorentz force law would then acquire an extra term

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) + \nabla(\mathbf{m}_i \cdot \mathbf{B}).$$

The magnetic field *could* do work on these “intrinsic” dipoles (which experience no motional or Faraday emf, since they enclose no flux). I don't know whether a consistent theory can be constructed in this way,²⁰ but in any event it is *not* classical electrodynamics, which is predicated on Ampère's assumption that *all* magnetic phenomena are due to electric charges in motion – and point magnetic dipoles must be interpreted as the limits of tiny current loops.

Of course, the *real* world *does* include particles with intrinsic spin, but they are governed by the laws of quantum mechanics, not classical mechanics. You can ask, if you like, whether *quantum* electrodynamics allows magnetic forces to do work. Unfortunately, this is not an easy question to address; “force” and “work” are not concepts that arise naturally in quantum mechanics. In *classical* electrodynamics, though, the answer is unequivocal: *magnetic forces do no work. Period.*

Problem 8.13. Derive Eq. 8.39. [*Hint:* Treat the lower loop as a magnetic dipole.]

Problem 8.14. Derive Eq. 8.43. [*Hint:* Use the method of Section 7.2.4, building the two currents up from zero to their final values.]

More Problems on Chapter 8

Problem 8.15.²¹ A very long solenoid of radius a , with n turns per unit length, carries a current I_s . Coaxial with the solenoid, at radius $b \gg a$, is a circular ring of wire, with resistance R . When the current in the solenoid is (gradually) decreased, a current I_r is induced in the ring.

(a) Calculate I_r , in terms of dI_s/dt .

²⁰ Semiclassical theories of intrinsic spin (and the associated intrinsic magnetic dipole moment) have been proposed from time to time – see, for example, J. Barandes, *Found. Phys.* **51**, 79 (2021). They tend to be cumbersome and implausible, but never mind: this is certainly something worth exploring.

²¹ For extensive discussion, see M. A. Heald, *Am. J. Phys.* **56**, 540 (1988).

- (b) Find the induced electric field as a function of s (the distance from the axis), and use this result to determine the work per unit time it does on the ring. Check that $P = I_r^2 R$.²²
- (c) The power delivered to the ring must come from the solenoid. Confirm this by calculating the Poynting vector just outside the solenoid (the *electric* field is due to the changing flux in the solenoid; the *magnetic* field is due to the current in the ring). Integrate over the entire surface of the solenoid, and check that you recover the correct total power.

Problem 8.16. An infinitely long cylindrical tube, of radius a , moves at constant speed v along its axis. It carries a net charge per unit length λ , uniformly distributed over its surface. Surrounding it, at radius b , is another cylinder, moving with the same velocity but carrying the opposite charge $(-\lambda)$. Find:

- (a) the energy per unit length stored in the fields;
 (b) the momentum per unit length in the fields;
 (c) the energy per unit time transported by the fields across a plane perpendicular to the cylinders.

Problem 8.17. A point charge q is located at the center of a toroidal coil of rectangular cross section, inner radius a , outer radius $a + w$, and height h , which carries a total of N tightly wound turns and current I .

- (a) Find the electromagnetic momentum $\mathbf{p}$ of this configuration, assuming that w and h are both much less than a (so you can ignore the variation of the fields over the cross section).
 (b) Now the current in the toroid is turned off, quickly enough that the point charge does not move appreciably as the magnetic field drops to zero. Show that the impulse imparted to q is equal to the momentum originally stored in the electromagnetic fields. [Hint: You might want to refer to Prob. 7.20.]

Problem 8.18.²³ A sphere of radius R carries a uniform polarization $\mathbf{P}$ and a uniform magnetization $\mathbf{M}$ (not necessarily in the same direction). Find the electromagnetic momentum of this configuration. [Answer: $(4/9)\pi\mu_0 R^3 (\mathbf{M} \times \mathbf{P})$.]

Problem 8.19.²⁴ Picture the electron as a uniformly charged spherical shell, with charge e and radius R , spinning at angular velocity ω .

- (a) Calculate the total energy contained in the electromagnetic fields.
 (b) Calculate the total angular momentum contained in the fields.
 (c) According to the Einstein formula ($E = mc^2$), the energy in the fields should contribute to the mass of the electron. Lorentz and others speculated that the *entire* mass of the electron might be accounted for in this way: $U_{\text{em}} = m_e c^2$. Suppose, moreover, that the electron's spin angular momentum is entirely attributable to the electromagnetic fields: $L_{\text{em}} = \hbar/2$.

²² This part was suggested by Sami Al-Jaber.

²³ For an interesting discussion and references, see R. H. Romer, *Am. J. Phys.* **63**, 777 (1995).

²⁴ See J. Higbie, *Am. J. Phys.* **56**, 378 (1988).

On these two assumptions, determine the radius and angular velocity of the electron. What is their product, ωR ? Does this classical model make sense?

Problem 8.20. Work out the formulas for u , $\mathbf{S}$, $\mathbf{g}$, and $\vec{\mathbf{T}}$ in the presence of magnetic charge. [Hint: Start with the generalized Maxwell equations (Eq. 7.53) and Lorentz force law (Eq. 8.44), and follow the derivations in Sections 8.1.2, 8.2.2, and 8.2.3.]

! **Problem 8.21.**²⁵ Suppose you had an electric charge q_e and a magnetic monopole q_m . The field of the electric charge is

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q_e}{r^2} \hat{\mathbf{r}}$$

(of course), and the field of the magnetic monopole is

$$\mathbf{B} = \frac{\mu_0}{4\pi} \frac{q_m}{r^2} \hat{\mathbf{r}}.$$

Find the total angular momentum stored in the fields, if the two charges are separated by a distance d . [Answer: $(\mu_0/4\pi)q_eq_m$.]²⁶ What does the conserved quantity $\mathbf{Q}$ in Prob. 5.48 represent, physically?

Problem 8.22. A magnetic dipole $\mathbf{m} = m\hat{\mathbf{z}}$ is at rest at the origin; an electric charge q_e is at rest at $\mathbf{r}$. Find the angular momentum in their fields. [Hint: The fastest way to get it is to exploit the result of Prob. 8.21.]

Problem 8.23. Consider a stationary ideal magnetic dipole $\mathbf{m}$ in a static electric field $\mathbf{E}$. Show that the fields carry momentum

$$\mathbf{p} = -\epsilon_0\mu_0(\mathbf{m} \times \mathbf{E}). \quad (8.45)$$

Here $\mathbf{E}$ is evaluated at the location of the dipole. We stipulate that the electric field is localized (it goes to zero at infinity), otherwise the momentum is ill defined (the integral does not converge). [Hint: There are several ways to do this. The simplest is to start with $\mathbf{p} = \epsilon_0 \int (\mathbf{E} \times \mathbf{B}) d\tau$, write $\mathbf{E} = -\nabla V$, and use integration by parts to show that

$$\mathbf{p} = \epsilon_0\mu_0 \int V \mathbf{J} d\tau. \quad (8.46)$$

²⁵ This system is known as **Thomson's dipole**. See I. Adawi, *Am. J. Phys.* **44**, 762 (1976) and *Phys. Rev.* **D31**, 3301 (1985), and K. R. Brownstein, *Am. J. Phys.* **57**, 420 (1989), for discussion and references.

²⁶ Note that this result is *independent of the separation distance d* ! It points from q_e toward q_m . In quantum mechanics, angular momentum comes in half-integer multiples of $\hbar$, so this result suggests that if magnetic monopoles exist, electric and magnetic charge must be quantized: $\mu_0 q_e q_m / 4\pi = n\hbar/2$, for $n = 1, 2, 3, \dots$, an idea first proposed by Dirac in 1931. If even *one* monopole is lurking somewhere in the Universe, this would “explain” why electric charge comes in discrete units. (However, see D. Singleton, *Am. J. Phys.* **66**, 697 (1998) for a cautionary note.)

So far, this is valid for *any* localized static configuration. For a current confined to an infinitesimal neighborhood of the origin we can approximate $V(\mathbf{r}) \approx V(\mathbf{0}) - \mathbf{E}(\mathbf{0}) \cdot \mathbf{r}$. Treat the dipole as a current loop, and use Eqs. 5.82 and 1.108.²⁷

Problem 8.24. Because the cylinders in Ex. 8.4 are left rotating (at angular velocities ω_a and ω_b , say), there is actually a residual magnetic field, and hence angular momentum in the fields, even after the current in the solenoid has been extinguished. If the cylinders are heavy, this correction will be negligible, but it is interesting to do the problem *without* making that assumption.²⁸

- Calculate (in terms of ω_a and ω_b) the final angular momentum in the fields. [Define $\boldsymbol{\omega} = \omega \hat{\mathbf{z}}$, so ω_a and ω_b could be positive or negative.]
- As the cylinders begin to rotate, their changing magnetic field induces an extra azimuthal electric field, which, in turn, will make an additional contribution to the torques. Find the resulting extra angular momentum, and compare it with your result in (a). [Answer: $-\mu_0 Q^2 \omega_b (b^2 - a^2)/4\pi I \hat{\mathbf{z}}$.]

Problem 8.25.²⁹ A point charge q is a distance $a > R$ from the axis of an infinite solenoid (radius R , n turns per unit length, current I). Find the linear momentum and the angular momentum (with respect to the origin) in the fields. (Put q on the x -axis, with the solenoid along z ; treat the solenoid as a nonconductor, so you don't need to worry about induced charges on its surface.) [Answer: $\mathbf{p} = (\mu_0 q n I R^2/2a) \hat{\mathbf{y}}$; $\mathbf{L} = \mathbf{0}$.] What if $a < R$?

! **Problem 8.26.** A point electric dipole $\mathbf{p}$ is located at the center of a spherical shell (radius R) carrying a surface current $\mathbf{K} = K_0 \sin \theta \hat{\boldsymbol{\phi}}$.

- Write down the electric and magnetic fields, inside and outside the sphere. [Hints: Don't forget the contact term in the electric field; for the magnetic field refer to Exs. 5.11 and 6.1, and express your answer in terms of the magnetic dipole moment $\mathbf{m}$ of the sphere.]
- Calculate the momentum in the fields.
- Suppose we turn off the electric field, by connecting a fine wire between the two ends of the dipole and allowing it to discharge quasistatically. Find the net impulse delivered to the current in the wire. [Hint: In parts (c) and (d) it may help to set the z -axis along $\mathbf{p}$, instead of $\mathbf{m}$, but always express your final answer in coordinate-independent form.]

²⁷ As it stands, Eq. 8.45 is valid only for *ideal* dipoles. But $\mathbf{g}$ is linear in $\mathbf{B}$, and therefore, if $\mathbf{E}$ is held fixed, obeys the superposition principle: for a *collection* of magnetic dipoles, the total momentum is the (vector) sum of the momenta for each one separately. In particular, if $\mathbf{E}$ is *uniform* over a localized steady current distribution, then Eq. 8.45 is valid for the whole thing, only now $\mathbf{m}$ is the *total* magnetic dipole moment.

²⁸ This problem was suggested by Paul DeYoung.

²⁹ See F. S. Johnson, B. L. Cragin, and R. R. Hodges, *Am. J. Phys.* **62**, 33 (1994), and B. Y.-K. Hu, *Eur. J. Phys.* **33**, 873 (2012) for discussion of this and related problems.

- (d) Meanwhile, the changing electric field induces a magnetic field, $\mathbf{B}_{\text{ind}}$; what is this field, and what impulse does it deliver to the sphere?
- (e) Suppose instead we turn off the magnetic field, by gradually decreasing K_0 . The changing magnetic field induces an electric field, $\mathbf{E}_{\text{ind}}$. What impulse does this deliver to $\mathbf{p}$?
- (f) Comment on your results. Is momentum conserved, here?³⁰

Problem 8.27.

- (a) Carry through the argument in Section 8.1.2, starting with Eq. 8.6, but using $\mathbf{J}_f$ in place of $\mathbf{J}$. Show that the Poynting vector becomes

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}, \quad (8.47)$$

and the rate of change of the energy density in the fields is

$$\frac{\partial u}{\partial t} = \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} + \mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t}.$$

For *linear* media, show that³¹

$$u = \frac{1}{2}(\mathbf{E} \cdot \mathbf{D} + \mathbf{B} \cdot \mathbf{H}). \quad (8.48)$$

- (b) In the same spirit, reproduce the argument in Section 8.2.2, starting with Eq. 8.14, with ρ_f and $\mathbf{J}_f$ in place of ρ and $\mathbf{J}$. Don't bother to construct the Maxwell stress tensor, but do show that the momentum density is³²

$$\mathbf{g} = \mathbf{D} \times \mathbf{B}. \quad (8.49)$$

Problem 8.28. A circular disk of radius R and mass M carries n point charges (q), attached at regular intervals around its rim. At time $t = 0$ the disk lies in the xy plane, with its center at the origin, and is rotating about the z -axis with angular velocity ω_0 , when it is released. The disk is immersed in a (time-independent) external magnetic field

$$\mathbf{B}(s, z) = k(-s\hat{\mathbf{s}} + 2z\hat{\mathbf{z}}),$$

where k is a constant.

- (a) Find the position of the center of the ring, $z(t)$, and its angular velocity, $\omega(t)$, as functions of time. (Ignore gravity.)
- (b) Describe the motion, and check that the total (kinetic) energy – translational plus rotational – is constant, confirming that the magnetic force does no work.³³

³⁰ This is not the end of the story, as we shall see in Chapter 12 (Prob. 12.43).

³¹ Refer to Section 4.4.3 for the meaning of “energy” in this context.

³² For over 100 years there has been a raging debate (still not completely resolved) as to whether the field momentum in polarizable/magnetizable media is Eq. 8.49 (Minkowski's candidate) or $\epsilon_0\mu_0(\mathbf{E} \times \mathbf{H})$ (Abraham's). See D. J. Griffiths, *Am. J. Phys.* **80**, 7 (2012).

³³ This cute problem is due to K. T. McDonald, <http://kirkmcd.princeton.edu/examples/disk.pdf> (who draws a somewhat different conclusion).

Problem 8.29. A particle with magnetic dipole moment $\mathbf{m} = m\hat{\mathbf{z}}$ and (positive) charge Q is fixed at the origin. A particle of (negative) charge q is released from rest at infinity, and “falls” in, attracted by Q and deflected by the magnetic field of $\mathbf{m}$. Its resulting trajectory is $\mathbf{r}(t) = (r(t), \theta(t), \phi(t))$.

(a) Show that the z component of the torque on q is

$$N_z = \frac{\mu_0 q m}{4\pi} \left(\frac{\sin^2 \theta}{r^2} \dot{r} - 2 \frac{\sin \theta \cos \theta}{r} \dot{\theta} \right) = -\frac{\mu_0 q m}{4\pi} \frac{d}{dt} \left(\frac{\sin^2 \theta}{r} \right),$$

and hence the z component of its angular momentum is

$$L_z = -\frac{\mu_0 q m}{4\pi} \frac{\sin^2 \theta}{r}.$$

(b) Where did this angular momentum come from? [*Hint:* Refer to Prob. 8.22.]